

Academic Review Report R2

Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting

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Review Standard: Strict (JBF Referee Standard)

Reference: R1 review dated 2026-04-05; K899 unified VaR experiment

R2 Review Summary

R1 Critical Issues resolved	1 of 5
R1 Critical Issues still open	4 of 5
New issues identified in R2	2 items
Overall Assessment	Major Revision (reduced scope)

Overall R2 impression. The v2 revision has successfully resolved the most damaging R1 issue—the Henriksson-Merton gamma contradiction (C1)—by explicitly labeling the two estimates as coming from different strategies and different sample periods. The revised text at Section 5.4 is now clear and internally consistent. However, four of the five original Critical issues remain unaddressed, and K899—a unified VaR experiment that was specifically designed to fix C3 (Table 5 cherry-picking)—has not yet been incorporated into the paper’s tables. The revision also introduces a new concern: the K899 results substantially differ from the numbers currently in the paper’s Table 5, creating a fresh traceability problem.

Status of R1 Critical Issues

C1: HM γ Internal Contradiction

RESOLVED

R1 finding: Section 4.7 reported $\hat{\gamma}_{HM} = -0.035$ ($t = -0.39$) while Section 5.4 reported $\hat{\gamma}_{HM} = -0.043$ ($t = -4.06$), creating an apparent contradiction for the same test.

R2 assessment: The revised body_v2.tex now clearly distinguishes:

- **Section 4.6** (line 376): Baseline 12/VIX strategy, full 2005–2026 sample. $\hat{\gamma}_{HM} = -0.035$, $t = -0.39$, $p = 0.70$. No timing.
- **Section 5.4** (line 436): Hybrid VT strategy, 2014–2026 subsample ($N \approx 3,100$). $\hat{\gamma}_{HM} = -0.043$, $t = -4.06$, $p < 0.001$. Significant negative.

The text explicitly states: “The significant negative γ_{HM} for Hybrid VT—contrasting with the insignificant -0.035 for the baseline 12/VIX strategy—reflects Hybrid VT’s more aggressive volatility scaling in the shorter, higher-volatility 2014–2026 subsample.” This is a clean resolution. The different magnitudes are explained by (a) different strategies (12/VIX vs. Hybrid), (b) different sample periods (21 years vs. 12 years), and (c) different volatility regimes covered. No further action needed.

Minor residual: The additions_jk.tex file (not included in main_v2.tex) still contains the old Section 4.7 text with $\hat{\gamma}_{HM} = -0.035$. Since it is no longer compiled, this is not a paper issue, but the file should be marked as superseded or removed to avoid future confusion.

C2: Table 5 Kupiec p -values Falsely Equalized

STILL OPEN

R1 finding: Both GJR+Student- t and GJR+HistSim reported Kupiec $p = 0.60$. Actual values from K802/K824v2 were 0.67 and 0.64 respectively.

R2 assessment: Table ?? in tables.tex (line 95–96) **still reports** $p = 0.60$ **for both rows**. This has not been corrected.

Compounding problem from K899: The new unified experiment K899 covers a much longer OOS period (2020–2025, $N = 1508$) compared to the paper’s 2023–2024 window ($N = 502$). K899 reports:

- GJR + Student- t : 23 violations (1.53%), Kupiec $p = 0.057$, Basel **Green**
- GJR + HistSim: 19 violations (1.26%), Kupiec $p = 0.330$, Basel **Yellow** (5 in rolling 250-day window)

These numbers differ dramatically from both the paper’s current Table 5 and the R1-flagged K802/K824v2 values, because K899 uses a 6-year evaluation window rather than 2 years. **The author must decide which OOS period to use and update Table 5 accordingly.** If the 2023–2024 window is retained, the p -values must be corrected to 0.67 and 0.64. If K899’s longer window is adopted, the entire table must be rewritten.

C3: Table 5 Cherry-Picks from Multiple Experiments

PARTIALLY

ADDRESSED

R1 finding: Table 5 mixed results from K799, K802, and K824v2 with different refit schedules, cherry-picking the most favorable count from each.

R2 assessment: K899 was explicitly designed to fix this (its title: “Unified VaR/ES for Paper 1, replaces cherry-picked Table 5”). K899 runs all seven VaR configurations with identical settings (expanding window, refit every 63 days, HS window = 500). However, **the paper’s Table 5 has not been updated to use K899 data**. The current table still shows the old mixed-source numbers.

Furthermore, K899 reveals important differences:

- K899 GJR+Normal: 34 violations (2.25%, Kupiec $p = 0.000$) vs. paper’s 10 violations (1.99%, $p = 0.049$)
- K899 GJR+Student- t : 23 violations (1.53%, $p = 0.057$) vs. paper’s 6 violations (1.20%, $p = 0.60$)
- K899 GJR+HistSim: 19 violations (1.26%, $p = 0.330$) vs. paper’s 4 violations (0.80%, $p = 0.60$)

The discrepancies are large and arise from K899 covering 6 years (including 2020 COVID) vs. the paper’s 2 years (2023–2024 only). The qualitative finding—that distributional upgrades improve VaR compliance—survives, but the specific numbers change substantially. Notably, under K899’s longer window, GJR+HistSim achieves the only ES pass (Acerbi-Szekely $z = 1.64$, $p = 0.10$), strengthening the paper’s narrative that HistSim is the most robust choice.

K899 also provides new ES data (Acerbi-Szekely test + Fissler-Ziegel scores), which directly addresses R1 Major Issue M5 (no Expected Shortfall analysis). Only GJR+HistSim passes the ES test. This is a significant finding that should be incorporated.

Action required:

1. Decide on OOS window: 2023–2024 (simpler narrative) or 2020–2025 (more powerful, addresses short-sample concern)
2. Rewrite Table 5 with K899 data for the chosen window
3. Add ES results (at minimum the Acerbi-Szekely pass/fail column)
4. Document K899 as the single source for all Table 5 values

C4: Table 1 Descriptive Statistics Untraceable

STILL OPEN

R1 finding: No experiment JSON backing Table 1’s values.

R2 assessment: No new experiment script or results JSON for Table 1 has been created. The descriptive statistics remain untraceable. This is the lowest-priority Critical among the five, but it still violates the reproducibility standard. Creating a simple `paper1_table1_descriptive.py` script would resolve it in under 30 minutes.

C5: Table 3 Non-SPY QLIKE Values Untraceable

STILL OPEN

R1 finding: Only SPY 2023–2024 QLIKE values were verified against K799/K802. The remaining 8 of 10 rows (QQQ, GLD, TLT, EEM, BTC across two periods) had no traceable source.

R2 assessment: No comprehensive cross-asset QLIKE experiment has been created. This remains a significant reproducibility gap for what is arguably the paper’s most important empirical table.

Status of R1 Major Issues

ID	R1 Issue	R2 Status	Assessment
M1	Paper length (62 pages)	UNCHANGED	body_v2.tex remains extensive. Complexity ceiling, HAR paradox, VIX sufficiency tests, and TZ momentum are all still in the main body rather than supplementary material.
M2	Abstract overclaims “6/6 OOS”	IMPROVED	Introduction now says “correctly classifying all nine DM comparisons” and adds the caveat “though the small cross-section ($N = 6$) limits statistical power.” The abstract-level overclaim is reduced.
M3	Proposition 1 mechanical confound	IMPROVED	The body now states “This correlation is partly mechanical” (line 468) and leads with the domain restriction. The framing is more honest. However, the $\rho = 1.000$ still appears prominently in the figure caption without the mechanical caveat.
M4	Harvey $t > 3.0$ inconsistency	UNCHANGED	The paper still reports some DM results with $p < 0.05$ as “significant” without noting they fail the Harvey threshold (e.g., K799 DM $t = -2.93$).

M5	No Expected Shortfall analysis	ADDRESSABLE	K899 now provides ES data (Acerbi-Szekely + Fissler-Ziegel). Only GJR+HistSim passes ES ($z = 1.64$, $p = 0.10$). All other methods fail ($p < 0.05$). This data exists but has not been incorporated into the paper.
M6	DeMiguel et al. (2024) mischaracterized	UNCHANGED	Line 46 still reads “hybrid implied-realized framework.” Should be “conditional multi-factor portfolio.”
M7	Hybrid VT Sharpe rounding (0.99)	UNCHANGED	Not checked in current tables.
M8	Missing references (Bekaert & Wu 2000, Figlewski & Wang 2000)	UNCHANGED	Neither reference has been added to the bibliography.

New Issues Identified in R2

N1 (NEW MAJOR): K899 Results Diverge Significantly from Paper’s Table 5

K899 was run over 2020–2025 ($N = 1508$), but Table 5 covers 2023–2024 ($N = 502$). The violation counts, rates, and Kupiec p -values differ by factors of 2–5 $\times$. If K899 is adopted as the source, Table 5’s narrative about “4 violations (0.80%)” for HistSim becomes “19 violations (1.26%),” and the “borderline failure” for GJR+Normal (10 violations) becomes a decisive failure (34 violations). The qualitative conclusions survive but the quantitative story changes substantially.

This is not a data error—it reflects the well-known sensitivity of VaR backtesting to the evaluation window. The longer window includes the extreme 2020 COVID period and the 2022 rate-hiking cycle, both of which stress-test the VaR models more severely. The longer window is arguably more informative but tells a less clean story.

Recommendation: Use K899 data but restrict to a well-motivated subperiod. Options:

1. **Full K899 (2020–2025):** Most powerful, includes multiple regimes, addresses the short-sample concern. But violation counts are high for all methods.
2. **K899 subset (2023–2025):** Matches the paper’s primary OOS better. Would need re-extraction from K899.
3. **Keep 2023–2024 but cite K899 for robustness:** Present the 502-day results in the main table, with a footnote or appendix showing the K899 6-year results confirm the qualitative ranking.

Option 3 is the most pragmatic: it preserves the clean narrative while adding the longer-sample evidence as robustness. But the p -values in the main table **must** be corrected regardless of which option is chosen.

N2 (NEW MAJOR): K899 ES Results Reveal a Starker Picture Than the Paper Acknowledges

K899’s Acerbi-Szekely ES backtest shows that **only GJR+HistSim passes** ($z = 1.64$, $p = 0.10$). All other methods—including GJR+Student- t ($z = 2.56$, $p = 0.01$), GJR+Skewed- t ($z = 2.32$,

$p = 0.02$), and both GARCH variants—fail the ES test. This is a dramatically stronger result than the VaR analysis alone: while Student- t and Skewed- t pass Kupiec, they systematically underestimate the magnitude of tail losses.

The Fissler-Ziegel joint VaR-ES scores confirm: GJR+HistSim achieves the best (least negative) score at both 1% and 5% levels, consistent with its zero-parameter design adapting to the empirical tail shape.

If the paper adds ES analysis (strongly recommended per R1-M5 and Basel III requirements), the HistSim dominance becomes even more compelling—**upgrading the paper’s practical recommendation from “HistSim is slightly better” to “HistSim is the only method that passes both VaR and ES.”** This would significantly strengthen the orthogonality narrative and the complexity ceiling argument.

Specific Verification: K899 Against Paper Numbers

Configuration		Paper Table 5	K899 (1508 days)	Match?	Note
GARCH	Normal viol.	7 (1.39%)	32 (2.12%)	No	Different period
GJR	Normal viol.	10 (1.99%)	34 (2.25%)	No	Different period
GJR	Student- t viol.	6 (1.20%)	23 (1.53%)	No	Different period
GJR	Student- t Kupiec p	0.60	0.057	No	Period + rounding
GJR	HistSim viol.	4 (0.80%)	19 (1.26%)	No	Different period
GJR	HistSim Kupiec p	0.60	0.330	No	Period + rounding
GJR	HistSim Basel	Green	Yellow	No	5 in 250-day window

All discrepancies arise from K899’s 6-year window vs. the paper’s 2-year window. The qualitative ranking (HistSim > Student- t > Normal) is preserved in K899.

R2 Revision Priority

Priority 1 (must fix before submission):

- Update Table 5 with corrected numbers.** Either: (a) rerun the 2023–2024 window from K899’s codebase to get unified results for the original period, then correct the Kupiec p -values to actual values; or (b) adopt K899’s 2020–2025 results and rewrite the surrounding narrative. Either way, all four rows must come from a single experiment with identical settings. *[Addresses C2, C3]*
- Add ES column to Table 5.** K899 provides Acerbi-Szekely results. At minimum, add a Pass/Fail column. The finding that only HistSim passes ES is the paper’s strongest practical result and directly addresses R1-M5 (no ES analysis) and the Basel III requirement. *[Addresses M5, N2]*

3. **Create traceable experiments for Tables 1 and 3.** Simple Python scripts that reproduce the exact numbers. Without these, the “replication code available upon request” claim in the title footnote is hollow. [Addresses C4, C5]

Priority 2 (strongly recommended):

4. **Fix DeMiguel et al. (2024) characterization** (line 46). Replace “hybrid implied-realized framework” with “conditional multifactor portfolio.” [Addresses M6]
5. **Add Bekaert and Wu (2000) and Figlewski and Wang (2000)** to the bibliography and cite appropriately in Section 5.1. [Addresses M8]
6. **Standardize Harvey threshold usage.** Either consistently apply $t > 3.0$ throughout (noting which results fail), or explicitly state when the standard 5% threshold is used and justify the choice. [Addresses M4]
7. **Move secondary analyses to supplementary material** to reduce paper length below 50 pages. Candidates: complexity ceiling details (Table 10), HAR paradox (Section 5.6), VIX sufficiency tests, TZ momentum appendix, diversification amplification. [Addresses M1]

Priority 3 (desirable):

8. Add a note to Figure ?? caption acknowledging the partly mechanical nature of $\rho = 1.000$. [Residual from M3]
9. Clean up or remove `additions_jk.tex` to avoid confusion with the superseded Section 4.7 text.
10. Report Hybrid VT Sharpe as 0.985 or 0.98 rather than 0.99. [Addresses M7]

What the Revision Did Well

1. **C1 resolution is exemplary.** The revised text clearly labels the two HM gamma estimates as coming from different strategies (12/VIX vs. Hybrid VT) and different periods (2005–2026 vs. 2014–2026), with an explicit sentence explaining why the magnitudes differ. This is exactly what R1 requested.
2. **OOS caveat strengthened.** The introduction now acknowledges that “the small cross-section ($N = 6$) limits statistical power” for the OOS classification test, and the 9/9 in-sample classification is correctly labeled as partly in-sample. The “6/6 correct OOS predictions” framing has been replaced with the more accurate “correctly classifying all nine DM comparisons.”
3. **Proposition 1 domain restriction is now explicit.** The text leads with the equity-type restriction ($\rho = 0.886$, $p = 0.019$, $N = 6$) and explicitly states that the mapping “does not extend to diverse asset classes” ($\rho = -0.448$). This is a substantial improvement in intellectual honesty.
4. **Overlapping-window discussion expanded.** Section 4.2.4 now includes HAC correction, block bootstrap CI, and non-overlapping 5-year window verification—a thorough three-pronged defense against the overlap concern.
5. **K899 experiment created.** The unified VaR/ES experiment with 7 methods, 1508 OOS days, both VaR and ES tests, and Fissler-Ziegel joint scores is exactly the kind of comprehensive, single-source experiment that R1 requested. It now needs to be *incorporated* into the paper.

Overall Assessment

Verdict: Major Revision (reduced scope).

The revision has addressed the most damaging issue (C1) and improved the framing of several overclaims (M2, M3). However, the traceability problems (C2–C5) remain open, and the new K899 experiment—while excellent—has not been integrated into the manuscript. The paper cannot be submitted until Table 5 is updated with unified data and the Kupiec p -values are corrected.

The good news is that all remaining issues are mechanical (table updates, script creation, reference additions) rather than conceptual. The core argument is sound. A focused revision of 1–2 days addressing Priority 1 items would bring the paper to a submittable state. Adding ES results from K899 would significantly strengthen the practical contribution.

Reviewer: Claude Opus 4.6 (IM context), acting as JBF referee. R2 review, 2026-04-05.

R1 reference: `paper/leverage-direction/reviews/review_r1.tex`

K899 source: `experiments/k899_unified_var_paper1_results.json`